Sparse and differentiable partial optimal transport plans through sliced approaches

Sliced optimal transport scales by projecting distributions onto random lines, where transport is solved in closed form; partial optimal transport relaxes mass conservation to compare distributions of unequal mass. Existing sliced partial methods only ever return a *distance* (or reweighted marginals), so the fact that different slices discard different mass never has to be reconciled — but this also means they produce no transport *plan* in the original space, and averaging the per-slice plans would destroy the sparsity that makes partial OT attractive. The internship aims at a sliced partial OT that yields a sparse, differentiable plan in the original space.

§1. Organization

- **Start date**: As soon as possible (flexible).
- **Localization**: IRISA, Rennes.
- **Supervision**:
  - Romain Tavenard (Prof., Université Rennes 2, MALT Inria team, IRISA, Rennes).
  - Laetitia Chapel (TAGADA research group, IRISA, Vannes).

§2. Context

Optimal transport (OT) provides a principled geometry for comparing probability distributions, but its cubic cost limits its use at scale. *Sliced* OT sidesteps this by projecting the distributions onto many random one-dimensional lines, solving the resulting transport problems in closed form via sorting, and averaging. *Partial* OT, in turn, relaxes the mass-conservation constraint so that only a fraction of the mass needs to be transported — valuable when distributions differ in total mass or contain outliers (Bonneel & Coeurjolly, 2019; Bai et al., 2023).

Two ingredients are now efficient: partial Wasserstein distances on the line admit a fast solution (Chapel & Tavenard, 2025), and existing sliced partial and unbalanced variants (Bai et al., 2023; Séjourné et al., 2023) are GPU-friendly. Crucially, though, these methods are designed as *distances*: they integrate the one-dimensional partial cost over projections, and what they recover in the original space is at most reweighted marginals (the inputs with outlier mass discarded), never a coupling. Séjourné et al. even note that per-slice mass selection makes the retained marginals *inconsistent across projections*, and sidestep it by reweighting the inputs globally. A separate line lifts a *single* optimized projection's one-dimensional plan back to the original space through a differentiable bilevel formulation (Chapel et al., 2025), yielding a sparse plan (a permutation) — but only in the *balanced* case.

§3. Scientific challenge

The two lines expose a gap. Averaging-based sliced partial OT never has to reconcile the different supports selected by different slices, precisely because it only computes a distance; but for the same reason it produces no plan in the original space, and naively averaging the per-slice plans would yield a *dense* coupling, destroying the sparse, interpretable matching that is the whole point of discarding outlier mass. Lifting a single projection's plan keeps that sparsity and is differentiable, but has only been done for balanced OT. The central question is therefore: **can a single-projector lift be extended to the partial setting, so that sliced partial OT returns a sparse, differentiable transport plan in the original space?** The obstacle is the mass-selection step: which mass a slice retains is a combinatorial choice, piecewise constant in the projection, so the lifted plan jumps as the projector moves. Quadratic or entropic relaxation of that selection (Nguyen et al., 2025) is one route to a usable gradient; whether it preserves sparsity and a valid approximation of the true partial plan is open.

The end goal is a sliced partial OT that returns a sparse, differentiable plan in the original space. Concretely, the candidate will:

1. formalize a partial one-dimensional plan and its lift to the original space, characterizing how the mass-selection step makes the lifted plan discontinuous in the projection direction;
2. extend the single-projector bilevel lift (Chapel et al., 2025) to the partial setting, using soft/entropic or quadratic relaxation (Nguyen et al., 2025) of the selection to obtain a differentiable yet sparse plan, and assess what approximation guarantees survive;
3. benchmark against exact (non-sliced) partial OT and against averaging-based sliced partial baselines (Bai et al., 2023; Séjourné et al., 2023) on problems small enough to admit ground truth, measuring plan sparsity, value, and gradient accuracy;
4. validate the estimator as a differentiable loss / plan in a downstream task (e.g. robust point-cloud matching or generative alignment), reporting sparsity, stability, and convergence.

Candidates should possess:

- good Python coding skills,
- knowledge of the fundamentals of optimal transport and numerical optimization,
- familiarity with the POT (Python Optimal Transport) library is a plus,
- ability to communicate in English in a scientific context.

Funding for a Ph.D. has already been secured: upon a successful internship, the candidate will be offered the opportunity to continue this work as a fully funded doctoral thesis. A successful candidate will develop or reinforce skills key to that continuation, namely:

- a practical and theoretical understanding of sliced and partial optimal transport, including transport plans and their lifts,
- the ability to turn a structural obstacle into a differentiable algorithm,
- experience benchmarking differentiable OT losses and plans for downstream training.

§4. Bibliography

Bai, Y., Schmitzer, B., Thorpe, M., & Kolouri, S. (2023). *Sliced Optimal Partial Transport.* CVPR.

Bonneel, N., & Coeurjolly, D. (2019). *SPOT: Sliced Partial Optimal Transport.* ACM Transactions on Graphics.

Chapel, L., & Tavenard, R. (2025). *One for All and All for One: Efficient Computation of Partial Wasserstein Distances on the Line.* ICLR.

Chapel, L., Tavenard, R., & Vaiter, S. (2025). *Differentiable Generalized Sliced Wasserstein Plans.* NeurIPS. arXiv:2505.22049.

Nguyen, K., et al. (2025). *Sparse Partial Optimal Transport via Quadratic Regularization.* arXiv:2508.08476.

Séjourné, T., Bonet, C., Fatras, K., Nadjahi, K., & Courty, N. (2023). *Unbalanced Optimal Transport meets Sliced-Wasserstein.* arXiv:2306.07176.